

Introduction to Big Data

USJ – FS – Academic Year 2025–2026 – Anthony Riachi

Chapter 1 – Introduction to Big Data

Section 1. Why and Where is Big Data

Data generated per minute



"2.1Million"



"3.8Million"



"1.0Million"



"4.5Million"



"188Million"

DATA IS EVERYWHERE

Evolution of Data Systems (Overview)

Data systems have evolved over time to handle:

- Increasing data volume
- Growing number of users
- Higher data complexity
- Global connectivity

Each evolution step was driven by **limitations of the previous model**.

Evolution 1: Centralized Computing

Data systems have evolved over time to handle:

- Increasing data volume
- Growing number of users
- Higher data complexity
- Global connectivity

Each evolution step was driven by **limitations of the previous model**.



CERN

Evolution 2: Distributed Computing (Local Networks)

To overcome limitations of centralized systems:

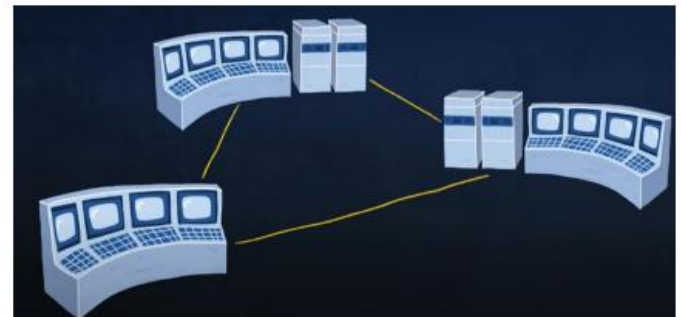
- Multiple machines were connected together
- Data and computation were shared across several computers
- Systems operated within the same organization or location

Improvements

- Better performance than a single machine
- Increased storage and computing power

Remaining limitations

- Limited geographic reach
- Still difficult to scale globally



Evolution 3: Distributed Systems Across Regions

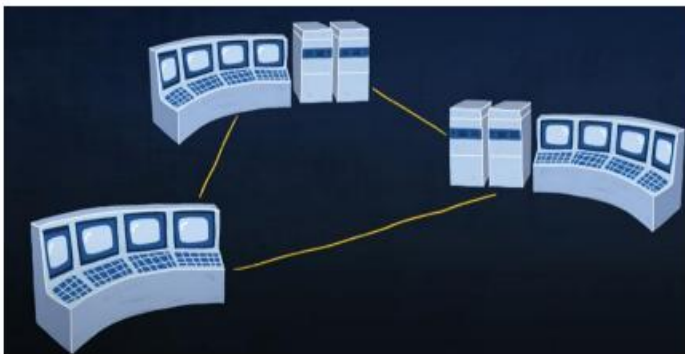
As data continued to grow:

- Systems expanded across **multiple regions**
- Data and services were replicated in different locations
- Communication between machines became essential

Benefits

- Higher availability
- Better fault tolerance
- Improved access for users in different regions

This step introduced the need for **data synchronization and coordination**.



Evolution 4: Global-Scale Distributed Systems (Big Data Era)

Modern systems operate at a global scale:

- Data is generated and processed **worldwide**
- Systems consist of **thousands of machines**
- Data is distributed across data centers and continents

Key characteristics

- Massive scale
- High fault tolerance
- Continuous data generation
- Need for distributed storage and processing frameworks

This evolution leads directly to **Big Data technologies**.

Evolution 4: Global-Scale Distributed Systems (Big Data Era)

- Data growth forced systems to evolve
- Single machines are no longer sufficient
- Distribution is not a choice, but a necessity
- Big Data frameworks exist to manage this complexity

This course focuses on **how modern systems handle data at scale.**



Other Applications

NETFLIX



Hurricane Sandy
in 2012



Section 2. What is Big Data?

What is Big Data?

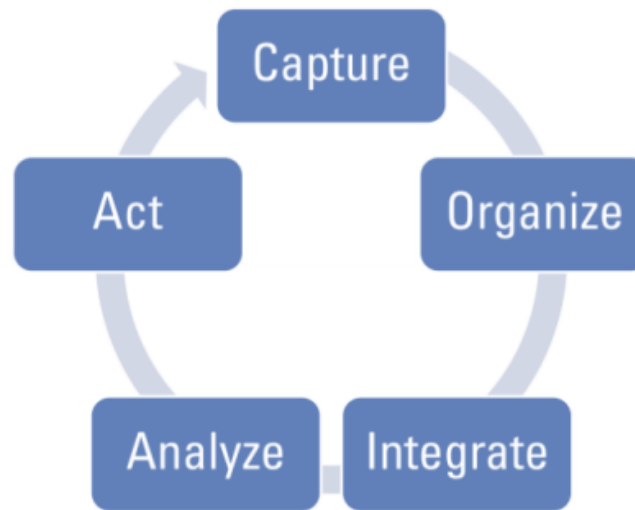
- Big Data refers to **digital information** that is **difficult to store, move, or analyze** using traditional systems.
- Big Data is **so large and complex** that it exceeds the capabilities of current technologies,
forcing us to develop **new storage and processing techniques**.
- In many cases, the **volume and complexity of data** exceed what a single computer can handle,
making processing **slow, inefficient, or impossible** without distributed systems.

Big Data Process (Overview)

Big Data follows a **continuous process** that transforms raw data into action.

The main steps are:

- Capture
- Organize
- Integrate
- Analyze
- Act



Each step depends on the previous one and supports decision-making.

Big Data Process (Capture)

What this step means

- Collecting raw data from different sources
- Data can be generated continuously or in large batches

Examples of data sources

- Sensors and IoT devices
- User activity on websites and applications
- Business transactions and logs

At this stage, data is **raw and unprocessed**.

Big Data Process (Organize)

What this step means

- Structuring and storing collected data
- Making data easier to access and manage

Key ideas

- Choosing how data is stored
- Handling large volumes efficiently
- Preparing data for further processing

This step focuses on **storage and structure**, not analysis.

Big Data Process (Integrate)

What this step means

- Combining data from multiple sources
- Ensuring data consistency and compatibility

Why it is important

- Data often comes in different formats
- Information must be unified before analysis

Integration allows data to be **used together meaningfully**.

Big Data Process (Analyze)

What this step means

- Processing data to extract insights and patterns
- Using algorithms, queries, or frameworks

Typical goals

- Detect trends
- Identify anomalies
- Support predictions and decisions

This is where **value is extracted from data**.

Big Data Process (Act)

What this step means

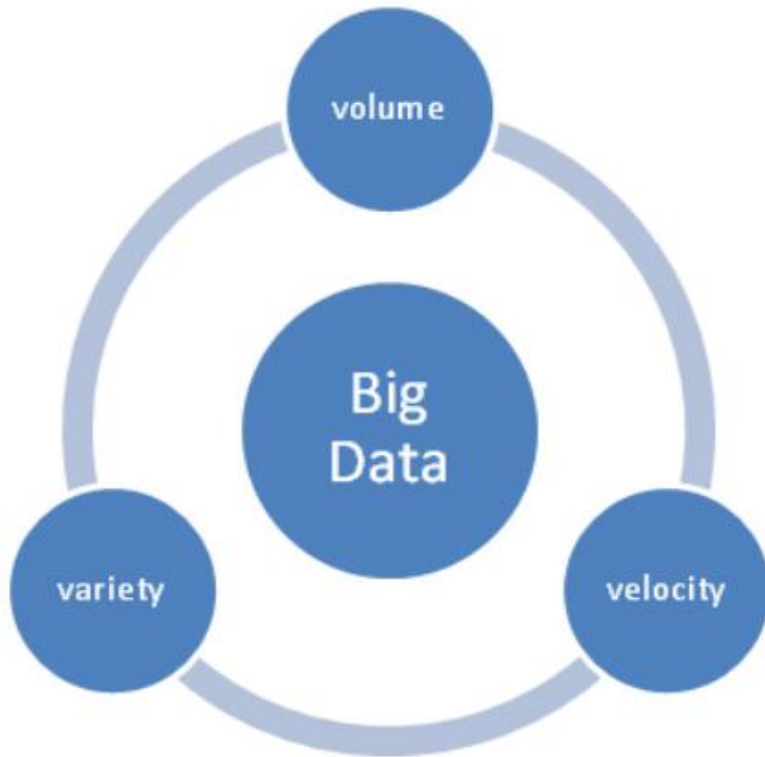
- Using analysis results to take action
- Supporting decisions or automated responses

Examples

- Business decisions
- System optimization
- Alerts or recommendations

This step closes the loop and may generate **new data**, restarting the process.

Big Data Characteristics



Big Data is characterized by the 3 Vs:

- **Volume**
- **Velocity**
- **Variety**

Big Data Characteristics - Volume

What Volume means

- Volume refers to the **amount of data** being generated and stored

Why it matters

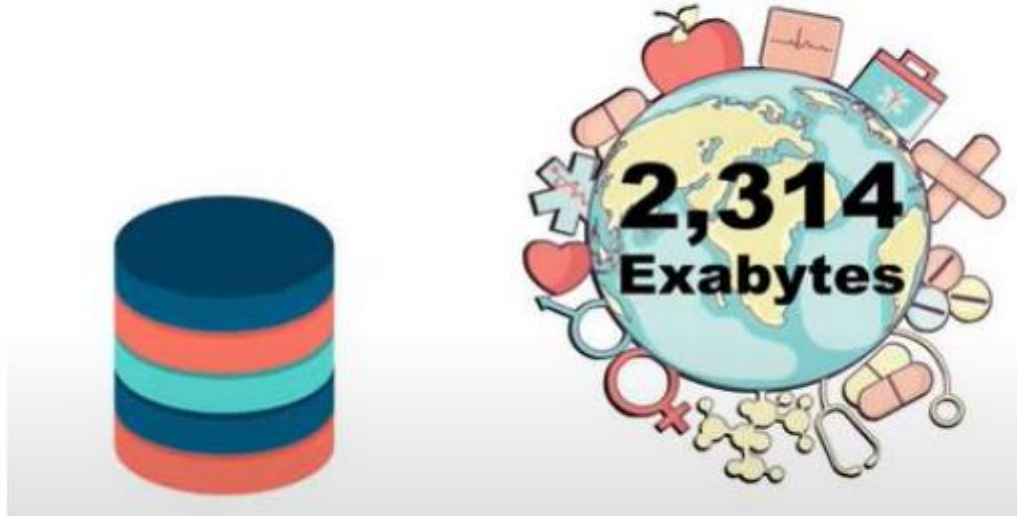
- Data sizes can reach **terabytes, petabytes, or more**
- A single computer cannot store or process this data efficiently

Example

- Years of transaction records or user activity logs

Large volume requires **distributed storage systems**.

Big Data Characteristics - Volume



Big Data Characteristics - Velocity

What Velocity means

- Velocity refers to the **speed at which data is generated and processed**

Why it matters

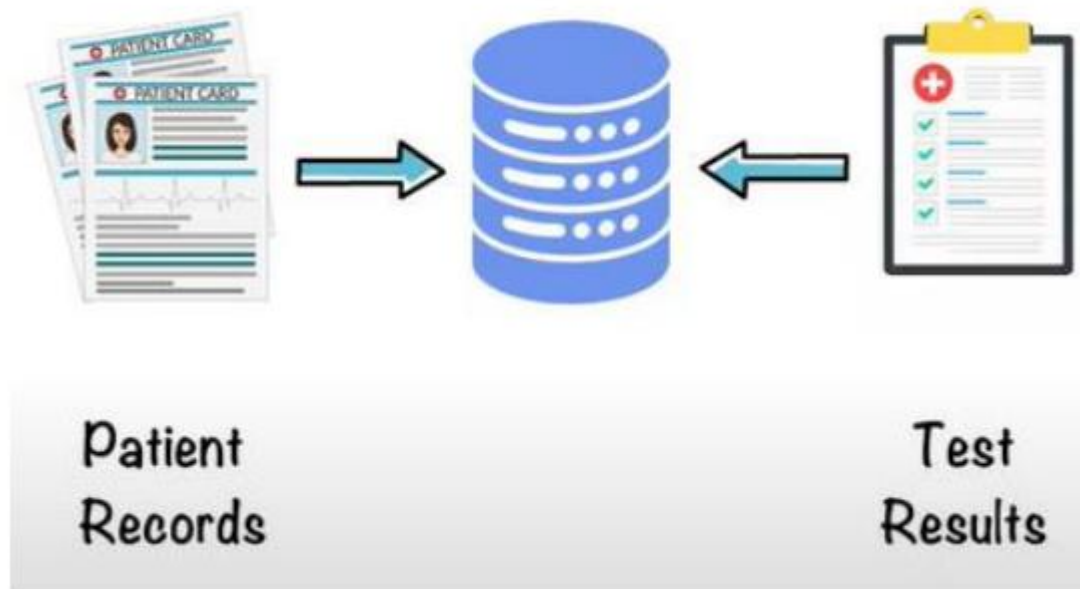
- Data may arrive continuously or in very large batches
- Systems must process data quickly to remain useful

Example

- Real-time user activity, sensor readings, or system logs

High velocity requires **efficient and scalable processing frameworks**.

Big Data Characteristics - Velocity



Big Data Characteristics - Variety

What Variety means

- Variety refers to the **different types and formats of data**

Types of data

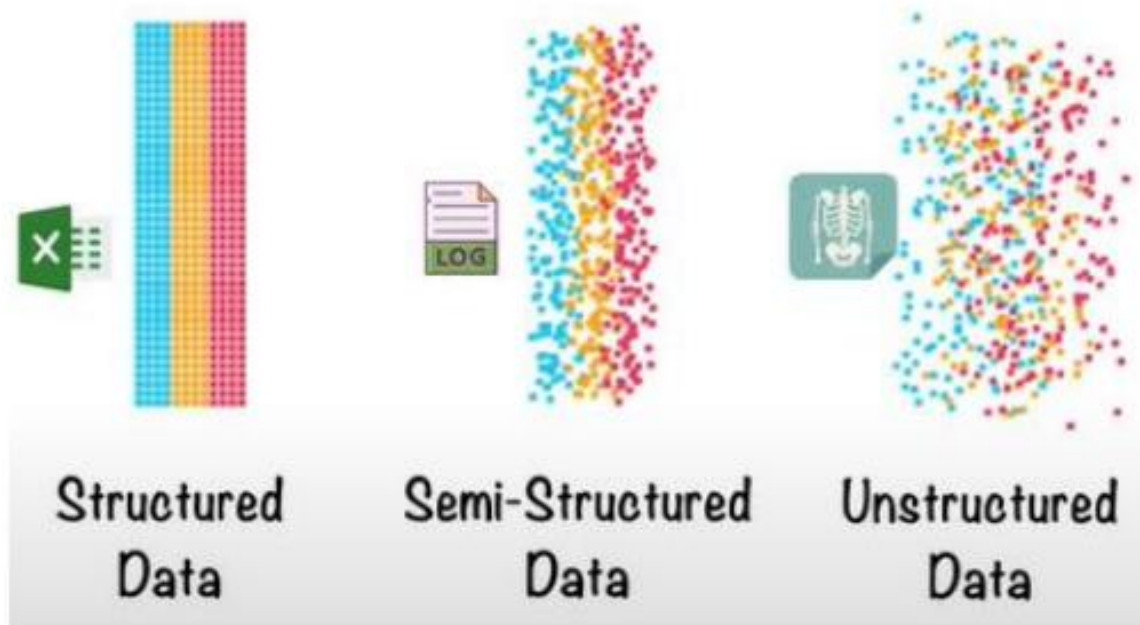
- Structured data (tables, databases)
- Semi-structured data (JSON, XML)
- Unstructured data (text, images, videos, logs)

Why it matters

- Data cannot always fit into a single table or schema

Variety requires **flexible storage and processing solutions**.

Big Data Characteristics - Variety



Section 3. Big Data Architecture

Big Data Architecture — Problem Analysis

Before designing a Big Data system, we must analyze the problem by answering key questions:

- What kind of data are we collecting?
- What are the functional requirements of the system?
- What level of performance is required (processing speed, computing power, latency)?
- Do we need redundancy to protect against failures, latency, or downtime (risk analysis and fault tolerance)?
- How precise does the data need to be?
- How much data must be stored today? How much in the future?



Answering these questions allows us to determine the **most suitable architecture** for our Big Data application.

Architecture — Additional Characteristics

Big Data architectures have specific characteristics due to the scale and diversity of data.

- Big Data systems collect **large volumes of data from multiple sources**
- As a result, **data integration APIs** are essential in Big Data systems
- **REST APIs** (Representational State Transfer) are commonly used for data exchange
- The architecture must be **robust and redundant** to handle failures
- Security is a **core requirement** of Big Data architectures
- Operational data sources must be **properly managed and monitored**

These characteristics ensure that Big Data systems are **scalable, reliable, and secure**.

Redundant Physical Infrastructure

Big Data systems rely on a **robust and distributed physical infrastructure**.

- Big Data requires a **robust physical infrastructure** to operate continuously
- Uses a **distributed computing model**, where:
 - Systems are spread across **multiple locations**
 - Components are connected through **networks or distributed file systems**
 - Extremely large volumes of data can be supported
- Big Data systems rely on **redundant solutions** to avoid failures:
 - Network redundancy
 - Storage and server redundancy
- **Examples of redundant infrastructures**
 - Private cloud environments
 - External cloud services
 - Software as a Service (SaaS) platforms



Redundancy ensures **availability, reliability, and fault tolerance**.

Redundant Physical Infrastructure

5 Factors to watch

Performance

- How fast and responsive the system should be ? “Make or Break” of the implementation

Availability

- How long can the business wait in case of a downtime ? Single point of failure ?

Scalability

- How big should the infrastructure be ? Become ?

Flexibility

- How easily can we modify modules, handle resources ?

Cost

- How much can we pay ? Do we need to compromise ?

Secure Infrastructure

Security must be **an integral part of Big Data systems**, not an afterthought.

- Security requirements must be considered **from the design phase**
- Big Data systems must secure data for:
 - Compliance
 - Privacy
 - Legal obligations
- User identity must be **verified and controlled**
- Threat detection mechanisms are required:
 - More data sources lead to **more security threats**
- **Key challenge**
 - Data encryption is essential, but it can **consume system resources**
 - A balance between security and performance is required



Security ensures **trust, data protection, and system integrity**.

Operational Data Sources

Big Data systems integrate data from **many different sources**.

- Data types:
 - Structured
 - Semi-structured
 - Unstructured
- Data is:
 - Continuously updated
 - Required to scale for many users
- Multiple database approaches may be used:
 - SQL databases
 - NoSQL databases (documents, graphs, etc.)
- There is **no single database solution**
- SQL and NoSQL systems are often **combined**

Designing operational data sources requires **flexibility and trade-offs**.

CAP Theorem

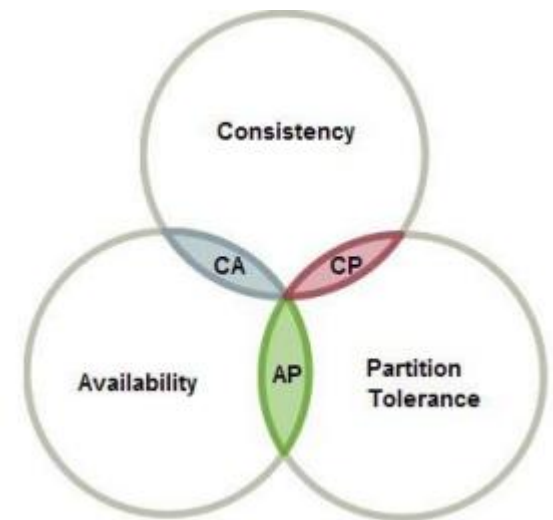
The **CAP Theorem** states that a distributed system can guarantee **only two out of three** properties:

- **Consistency**
- **Availability**
- **Partition Tolerance**

In Big Data systems:

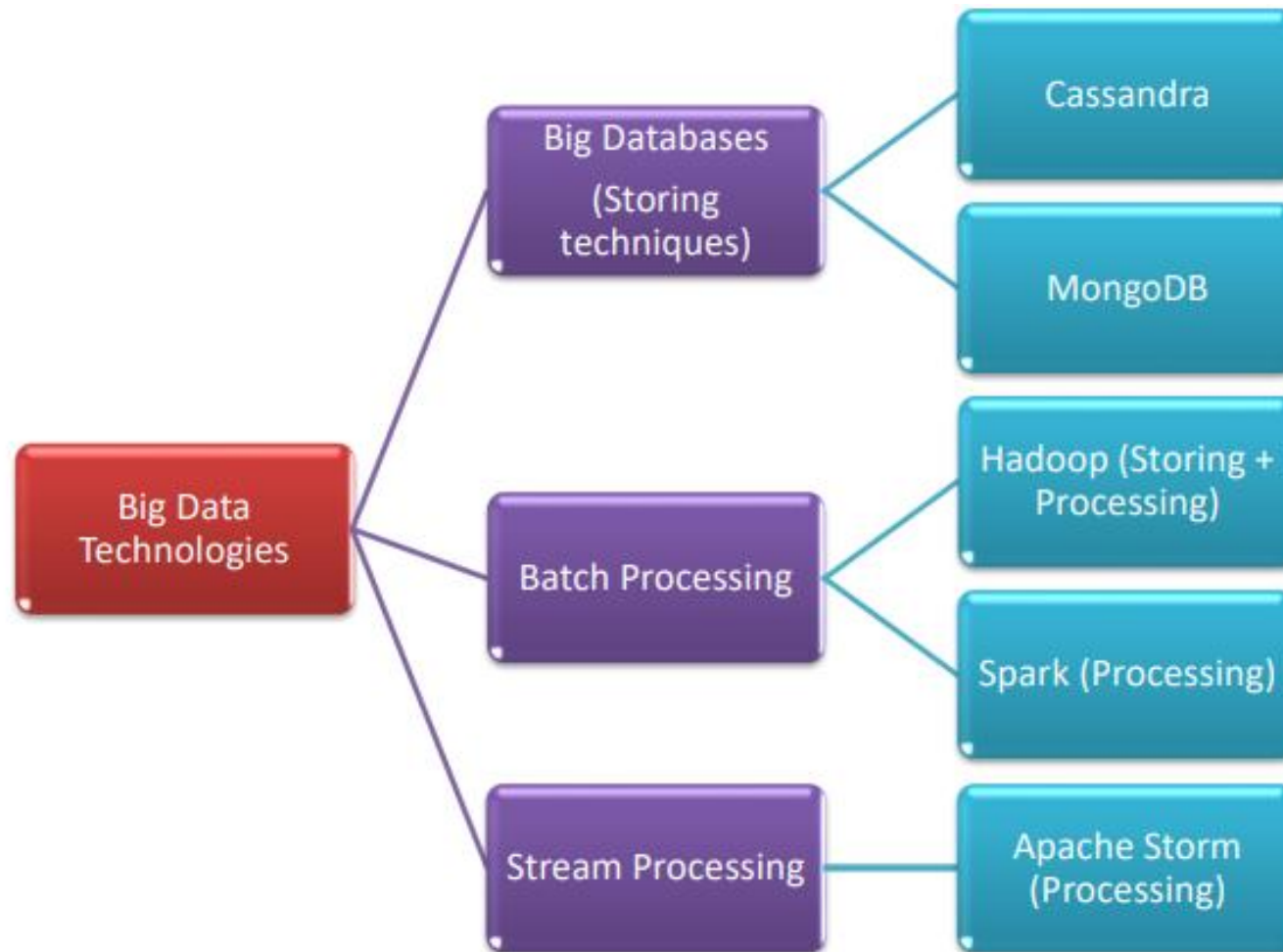
- Network failures **will happen**
- Trade-offs between these properties are **unavoidable**

System design depends on the **use case and priorities**.

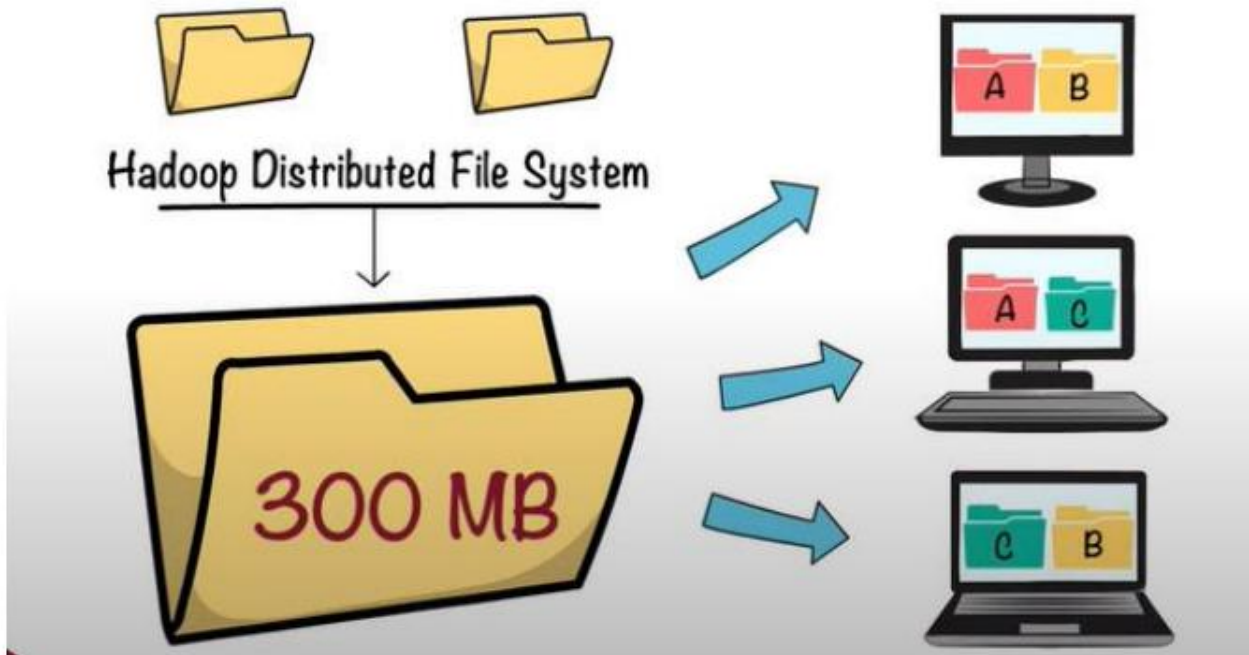


Section 4. Big Data frameworks example

Big Data Technologies & Frameworks

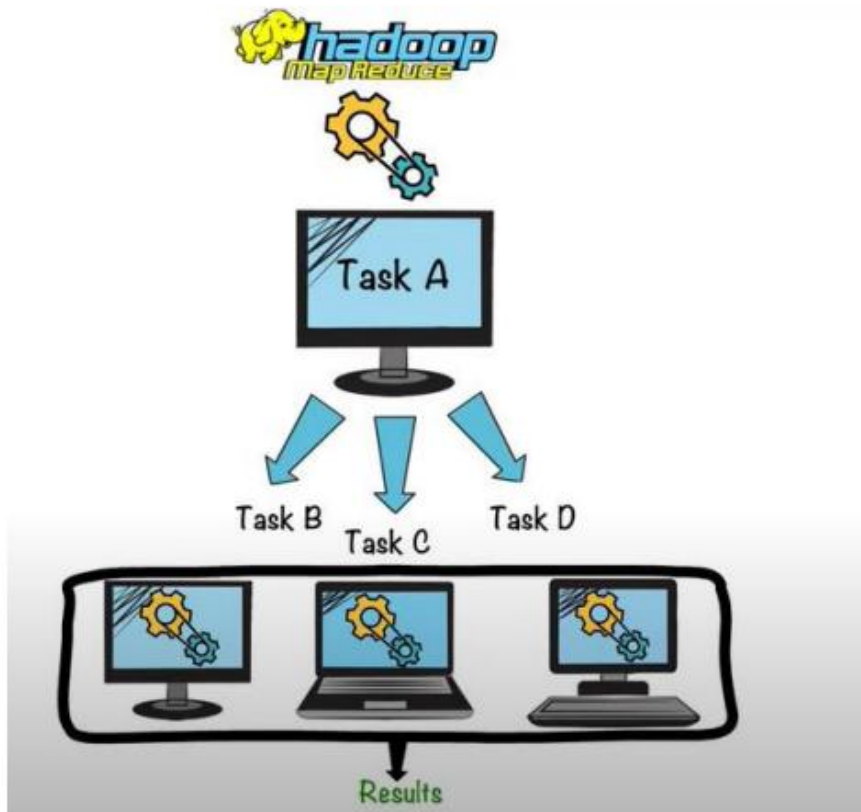


Hadoop Example - HDFS



**HDFS optimizes
the way data is
being stored**

Hadoop Example - Map Reduce



Hadoop Map Reduce optimizes the way data is being processed

Section 5. Recap Exercises

Recap Exercise

- In which of the below cases, Big Data may not be a good approach ?
 - Personal banking information
 - Health mobile applications (tracking steps etc.)
 - Grocery store sales
 - Agriculture supervision

Recap Exercise

- **How does Hadoop technology ensure redundancy ?**